Infinitely many cyclic Haagerup–Izumi fusion categories

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Abstract

We construct complex spherical fusion categories with cyclic Haagerup–Izumi fusion rules at every odd order $n \geq 3$ for which -1 belongs to the subgroup generated by 2 in $(\mathbb{Z}/n\mathbb{Z})^\times$. In particular, the orders $n = 2^m + 1$, $m \geq 1$, give infinitely many distinct ranks. The construction starts with explicit double-sine samples. A finite Fourier evaluation proves the quadratic identities; a contour difference and induction prove the ordered cubic identities. An unramified convolution lift supplies the exact field automorphism needed to treat the remaining indices. Algebraic reconstruction and Galois conjugation then give spherical and pseudo-unitary categorifications, respectively.

AI-use disclosure. AI systems were used to conduct the numerical searches, including writing and running computational code, to develop the proof arguments, and to write this preliminary LaTeX draft.

1 Statement and structure of the proof

For $G = \mathbb{Z}/n\mathbb{Z}$, the cyclic Haagerup–Izumi based ring has basis α^a, ρ_a $(a \in G)$ and multiplication

$$\alpha^a\alpha^b = \alpha^{a+b}, \qquad \alpha^a\rho_b = \rho_{a+b}, \qquad \rho_a\alpha^b = \rho_{a-b}, \tag{1.1}$$

$$\rho_a\rho_b = \alpha^{a-b} + \sum_{c\in G} \rho_c. \tag{1.2}$$

Its noninvertible Frobenius–Perron dimension is $d = (n + \sqrt{n^2+4})/2$.

Theorem 1.1. *Let $n \geq 3$ be odd. Suppose that $2^m \equiv -1 \pmod{n}$ for some $m \geq 1$. There exists a complex spherical fusion category with fusion rules (1.1)–(1.2) and noninvertible categorical dimension $\delta = -1/d$. There also exists a pseudo-unitary categorification of the same based ring. Consequently there are infinitely many pairwise inequivalent cyclic Haagerup–Izumi fusion categories.*

The argument below is independent of any conjectural identification of the coefficient fields with ray class fields or of the samples with Stark units. All finite sums and polynomial identities in the proof are exact. Numerical verification is recorded separately and is not used as a proof of an identity for unbounded n.

Throughout, A, B denote positive real periods, while A, B denote matrices. All matrix and phase indices are in G unless a signed integer representative is expressly chosen. The Kronecker symbol is written $\mathbf{1}_{\{\cdot\}}$, to distinguish it from the number δ.

2 The analytic candidate and its quadratic identities

Fix an odd integer $n \geq 3$, and put

$$d = \frac{n + \sqrt{n^2+4}}{2}, \qquad \delta = -d^{-1}, \qquad \tau = \delta - 1, \qquad A = d + 1, \qquad B = 1 + d^{-1}.$$

Then

$$\delta^2 = n\delta + 1, \qquad A - B = n, \qquad \Omega := A + B = AB, \qquad A/B = d, \qquad \tau = -B.$$

Let $\gamma(z) = \gamma^{(2)}(z; A, B)$ be the hyperbolic gamma function in the normalization of [1]. Thus

$$\gamma(z + A) = 2\sin(\pi z/B)\gamma(z), \qquad \gamma(z + B) = 2\sin(\pi z/A)\gamma(z), \tag{2.1}$$

$$\gamma(z)\gamma(\Omega - z) = 1, \qquad \operatorname*{Res}_{z=0} \gamma(z) = \frac{\sqrt{\Omega}}{2\pi}. \tag{2.2}$$

Its poles are $-rA - sB$, $r, s \geq 0$; its zeros are $\Omega + rA + sB$. It is positive on $(0, \Omega)$. Write $F(z) = \gamma(B + z)^{-1}$. The preceding identities give

$$F(0) = \sqrt{d}, \qquad F(n - z) = F(z)^{-1}, \qquad F(-a) = \frac{(-1)^{a+1}}{F(a)} \quad (a \in \mathbb{Z} \setminus \{0\}).$$

Define real phases, indexed by $\mathbb{Z}_n$, by

$$W_0 = -1, \qquad W_a = -(-1)^{a(a+n)/2}\frac{F(a)}{\sqrt{d}} \quad (1 \leq a < n). \tag{2.3}$$

In particular $W_a W_{-a} = \delta$ for $a \neq 0$. Define the core and its rescaling by

$$\mathsf{B}_{a,b} = \begin{cases} \delta - 2, & a = b = 0, \\ -1, & a = 0 \text{ or } b = 0 \text{ or } a = b, \quad (a,b) \neq (0,0), \\ W_a W_{b-a}/W_b, & \text{otherwise}, \end{cases} \qquad \mathsf{A}_{a,b} = \tau^{-1}\mathsf{B}_{a,b}. \tag{2.4}$$

For $b \neq 0$ the phase quotient in (2.4) applies also at $a = 0, b$. The inverse norms imply the order-three symmetry $\mathsf{B}_{a,b} = \mathsf{B}_{b-a,-a}$.

Proposition 2.1. *The matrix* (2.4) *satisfies*

$$\sum_{a \in \mathbb{Z}_n} \mathsf{A}_{g+a,h}\mathsf{A}_{h,a} = \mathbf{1}_{g=0} - \delta^{-1}\mathbf{1}_{h=0} \qquad (g, h \in \mathbb{Z}_n). \tag{2.5}$$

More precisely, for $0 < h < n$, *put*

$$\widehat{\mathsf{B}}_h(k) = \sum_{a=0}^{n-1} \mathsf{B}_{a,h} e^{-2\pi i k a/n}, \qquad \omega \equiv \frac{2\pi k}{n} + \pi(h-1) \pmod{2\pi}, \quad -\pi \leq \omega \leq \pi.$$

Then

$$\widehat{\mathsf{B}}_h(k) = -Be^{-ih\omega/2}\gamma\left(\frac{\Omega + h}{2} - \frac{\Omega\omega}{2\pi}\right)\gamma\left(\frac{\Omega + h}{2} + \frac{\Omega\omega}{2\pi}\right). \tag{2.6}$$

Proof. Set $R_h(z) = F(z)/F(z-h)$ and $K_h(z) = \pi e^{-i\omega z} R_h(z)/\sin \pi z$. At the integer samples, including $a = 0, h$,

$$\mathsf{B}_{a,h} + B\mathbf{1}_{a=h} = \frac{(-1)^{h(a+1)}}{\sqrt{d}\, F(h)} R_h(a). \tag{2.7}$$

Put $\alpha = 1/A$ and $\beta = 1/B$, so $\alpha + \beta = 1$. The period shifts and the elementary identity

$$(-1)^h \sin(\pi\alpha z) \sin(\pi\beta(z-h)) - \sin(\pi\beta z) \sin(\pi\alpha(z-h)) = \sin(\pi\alpha h) \sin \pi z$$

give

$$K_h(z+n) - K_h(z) = -4\pi \sin(\pi h/A) e^{-i\omega z} \gamma(-z)\gamma(z-h). \tag{2.8}$$

Let $L = \lfloor h/B \rfloor$ and choose $\sigma = -\varepsilon$, where $0 < \varepsilon < \min\{1, (L+1)B - h\}$. In the strip $\sigma < \operatorname{Re} z < \sigma + n$, the poles of K_h are $0, \dots, n-1$ and $p_\ell = h - \ell B$, $1 \le \ell \le L$. There are no numerator poles because the first pole of F is at $A > n$. The irrationality of B excludes coincidences, and $K_h(z+n)$ is regular at the additional poles. The standard gamma asymptotic gives, uniformly on this finite strip,

$$|R_h(x+it)| = O(e^{-\pi h|t|/\Omega}).$$

Thus the horizontal sides of the residue rectangle vanish, including when $|\omega| = \pi$.

Write $p_0 = h$ and

$$r_\ell = \operatorname*{Res}_{z=p_\ell} e^{-i\omega z} \gamma(-z)\gamma(z-h), \qquad I_\sigma = \int_{\sigma - i\infty}^{\sigma + i\infty} e^{-i\omega z} \gamma(-z)\gamma(z-h) \frac{dz}{i}.$$

The rectangle and (2.8) give

$$\sum_{a=0}^{n-1} \operatorname*{Res}_{z=a} K_h(z) = -2\sin(\pi h/A) I_\sigma - 4\pi \sin(\pi h/A) \sum_{\ell=1}^{L} r_\ell. \tag{2.9}$$

The two-gamma Fourier integral [1, Eq. (2.17)], with parameters $f = -h$, the other shift zero, and $\alpha_{\mathrm{BSS}} = (\Omega + h - \Omega\omega/\pi)/2$, states that

$$I_{\mathrm{sep}} = \sqrt{\Omega}\, e^{-ih\omega/2} \gamma(-h) \gamma\left(\frac{\Omega + h}{2} - \frac{\Omega\omega}{2\pi}\right) \gamma\left(\frac{\Omega + h}{2} + \frac{\Omega\omega}{2\pi}\right), \tag{2.10}$$

$$I_{\mathrm{sep}} = I_\sigma + 2\pi \sum_{\ell=0}^{L} r_\ell. \tag{2.11}$$

Here the separating contour is continued meromorphically from negative h and $|\omega| < \pi$. The second equality lists every pole that has crossed the vertical line. At the final integer h, no opposite poles collide: $h < n < A$ cannot equal $rA + sB$ with $r, s \ge 0$. The endpoint frequencies follow by continuity in the convergent positive-h regime.

Substituting (2.11) into (2.9) cancels the residues with $\ell \ge 1$. The remaining residue is

$$r_0 = \frac{\sqrt{\Omega}}{2\pi} e^{-i\omega h} \gamma(-h).$$

Now use

$$2\sin(\pi h/A)\gamma(-h) = (-1)^h F(h), \qquad \frac{\sqrt{\Omega}}{\sqrt{d}} = B.$$

Multiplication of the integer residue sum by $(-1)^h/(\sqrt{d}F(h))$ yields the right side of (2.6) plus $Be^{-i\omega h}$. Since $e^{-i\omega h} = e^{-2\pi i k h/n}$, this extra term cancels exactly the correction in (2.7).

For completeness, reflection of the evaluated gamma factors gives

$$\widehat{\mathsf{B}}_h(k)\widehat{\mathsf{B}}_{n-h}(-k) = B^2 e^{-2\pi i k h/n}. \tag{2.12}$$

Here is the phase check. By exchanging $h, n-h$ and conjugating, it suffices to take h odd and $0 \le k \le (n-1)/2$. Put $\lambda_k = \Omega k/n$. The first gamma arguments are $x_\pm = (\Omega + h)/2 \pm \lambda_k$, and the other two are $y_1 = \Omega + (n-h)/2 - \lambda_k$ and $y_2 = (n-h)/2 + \lambda_k$. Since $x_+ + y_1 = \Omega + A$ and $x_- + y_2 = A$, their product reduces to

$$\frac{\sin\bigl(\pi[(h-n)/(2B) + Ak/n]\bigr)}{\sin\bigl(\pi[(\Omega+h)/(2A) - Bk/n]\bigr)} = (-1)^{(h-n)/2+k}.$$

The two arguments divided by π sum to $(h-n+2)/2 + k$; the denominator argument lies strictly between zero and one. The exponential factors give (2.12). Finally $\mathsf{B}_{h,a} = \mathsf{B}_{a-h,n-h}$, so Fourier inversion of (2.12) proves (2.5) for $h \neq 0$. For $h = 0$, it follows directly from the fixed axes and $\delta^2 = n\delta + 1$. □

Lemma 2.2. *Every phase W_a, and every sampled value $F(a)$ with $0 \le a < n$, is algebraic over $\mathbb{Q}$.*

Proof. This uses only reality, the inverse norms, and one quadratic column. Put $q = (n+1)/2$, $u_a = \mathsf{B}_{a,1}$, and $v_a = \mathsf{B}_{1,a}$. Then

$$u_0 = u_1 = v_0 = v_1 = -1, \qquad u_a = u_{1-a}, \qquad v_a = \delta/u_a \quad (a \neq 0, 1).$$

Let $x_j = u_{q+j}$ and $y_j = v_{q+j}$. Both sequences are even. Their symmetric circulants T and V satisfy $TV = \tau^2 I$, by (2.5) at $h = 1$. If c_j are the coefficients of T^{-1}, then $c_{a-q} = v_a/\tau^2$.

On the real affine space $u_a = u_{1-a}$, $u_0 = u_1 = -1$, consider, locally where every free coordinate and $\det T$ are nonzero,

$$\Phi(u) = \log|\det T| + \eta \sum_{a=2}^{n-1} \log|u_a|, \qquad \eta = -\frac{n\delta}{\tau^2} > 0.$$

For a free reflection orbit represented by a, its derivative is the orbit multiplicity times $nc_{a-q} + \eta/u_a = 0$. Thus the given u is critical. Every directional derivative H of T is again a real symmetric circulant, so it commutes with T^{-1}, and

$$D^2 \log|\det T|[H, H] = -\operatorname{Tr}((T^{-1}H)^2) \le 0.$$

The coordinate logarithms have strictly negative Hessian on the free coordinates. Consequently the critical point is nondegenerate. The critical equations are rational over $\mathbb{Q}(\delta)$; clearing their nonzero denominators gives an isolated simple complex algebraic point. Hence each u_a is algebraic.

Put $R_a = \delta/u_a$ for $2 \le a < n$. The phase definitions imply

$$W_1^n = \delta \prod_{a=2}^{n-1} R_a, \qquad W_a = \frac{W_1^a}{\prod_{j=2}^{a} R_j} \quad (2 \le a < n).$$

They make every W_a algebraic; (2.3) then does the same for $F(a)$. □

3 Convolution and an exact Galois symmetry

Proposition 3.1. *Let $K = \mathbb{Q}(\delta)$ and $E = K(W_0, \dots, W_{n-1})$. Every phase is a unit at every prime above 2, the extension E/K is unramified there, and E has a K-automorphism σ satisfying*

$$\sigma(W_a) = W_{2a} \qquad (a \in \mathbb{Z}_n). \tag{3.1}$$

Proof. First, the zero Fourier mode gives a useful exact identity. For $0 < h < n$ let $r \in \{1, \dots, n-1\}$ represent $h/2$ modulo n. If h is odd, $r = (n+h)/2$ and both gamma arguments in (2.6) at $k = 0$ equal $B + r$. If h is even, $r = h/2$ and the arguments are $r, \Omega + r$; their product is $(-1)^r/F(r)^2$, with this sign canceled by the exponential factor. In both cases,

$$\sum_a \mathsf{B}_{a,h} = -B/F(r)^2.$$

Since $F(r)^2 = dW_r^2$ and $\mathsf{B}_{a,h} = W_a W_{h-a}/W_h$ for $h \neq 0$, this is

$$\sum_a W_a W_{2r-a} = -\delta\tau \, \frac{W_{2r}}{W_r^2} \qquad (r \in \mathbb{Z}_n). \tag{3.2}$$

The case $r = 0$ follows separately from $1 + (n-1)\delta = \delta(\delta - 1)$ and $W_0 = -1$.

Fix a valuation v above 2. Both δ and τ are units: the reduction of $X^2 - nX - 1$ is $X^2 + X + 1$. If $m = \min_a v(W_a) < 0$, choose r attaining the minimum. Pair the summands in (3.2) under $a \leftrightarrow 2r - a$. Because n is odd, the unique fixed summand is W_r^2; every other pair has valuation at least $v(2) + 2m > 2m$. Thus the left side has valuation $2m$, forcing $v(W_{2r}) = 4m < m$, a contradiction. Every phase is integral; the inverse norms make all phases units. Reduction now gives

$$\overline{W}_{2r} = \overline{W}_r^{\,4}. \tag{3.3}$$

The completion K_2 is the unramified quadratic extension of $\mathbb{Q}_2$. Put $\lambda = \delta\tau$ and consider the polynomial system

$$\mathcal{H}_r(X) = X_r^2 \sum_a X_a X_{2r-a} + \lambda X_{2r} = 0.$$

Its reduction is $X_r^4 + X_{2r} = 0$, with permutation-matrix Jacobian. A direct form of its Hensel lift is

$$(\mathcal{T}X)_{2r} = -\lambda^{-1} X_r^2 \sum_a X_a X_{2r-a}.$$

Choose a complete unramified integer ring containing the prescribed residue tuple; such a ring exists by the finite étale correspondence for henselian rings [3, Tag 04GK]. This map preserves the tuple and improves $X - Y \in 2^s\mathcal{O}^n$ to $\mathcal{T}X - \mathcal{T}Y \in 2^{s+1}\mathcal{O}^n$ for $s \geq 1$. Indeed, writing $X = Y + e$, the convolution difference is $2Y * e + e * e$ and the square difference has the same divisibility. Iteration therefore gives a unique integral lift in that unramified ring.

We also need uniqueness against ramified extensions. Normalize $v(2) = 1$, and suppose two integral tuples have the same residue, with $s = \min_a v(X_a - Y_a) > 0$. The same computation gives

$$\min_a v((\mathcal{T}X)_a - (\mathcal{T}Y)_a) \geq \min\{s+1, 2s\} > s.$$

Consequently two distinct fixed points cannot have the same residues. The completed analytic phases are fixed points by (3.2); they therefore coincide with their unramified lift. This proves unramifiedness of E/K at every prime above 2.

At any such completion let φ be arithmetic Frobenius over K_2, with residue action $x \mapsto x^4$. Both tuples $(\varphi(W_a))_a$ and $(W_{2a})_a$ solve the convolution system: the first because φ fixes λ, the second by reindexing. Their residues agree by (3.3). Uniqueness proves $\varphi(W_a) = W_{2a}$. Since all these images belong to E and doubling permutes the indices, restriction gives the automorphism (3.1). □

In particular, whenever $2^m \equiv -1 \pmod{n}$, the automorphism σ^m fixes δ and sends W_a to W_{-a} and $\mathsf{B}_{a,b}$ to $\mathsf{B}_{-a,-b}$. Thus simultaneous negation of indices preserves the vanishing of every polynomial residual with coefficients in K.

4 Product Fourier transforms and their contour corrections

Write $u_p(a) = \mathsf{B}_{a,p}$, with subscripts in $\mathbb{Z}/n\mathbb{Z}$, and set

$$S_{p,q,s}(k) = \sum_{a \bmod n} u_p(a)u_q(a+s)e^{-2\pi ika/n}.$$

For $g, h, g+h \neq 0$, let

$$\mathcal{C}_{g,h}(k,l) = \sum_m \mathsf{B}_{m,g+h}\mathsf{B}_{g,m+k}\mathsf{B}_{h,m+l} - \tau\mathsf{B}_{g+l,k}\mathsf{B}_{h+k,l}.$$

The order-three symmetry gives $\mathsf{B}_{g,a} = u_{-g}(a-g)$. Fourier transformation in k, followed in the second term by $a = k-g-l$, therefore gives the exact finite identity

$$\sum_k \mathcal{C}_{g,h}(k,l)e^{-2\pi ijk/n} = e^{-2\pi ijg/n}\left[\widehat{u}_{-g}(j)S_{g+h,-h,l-h}(-j) - \tau e^{-2\pi ijl/n}S_{-g-l,l,g+h+l}(j)\right]. \tag{4.1}$$

This reindexing uses no quadratic or cubic equation.

For signed representatives $-n < q < 0$, the column interpolation takes the particularly useful form

$$u_q(a) = \frac{(-1)^{q(a+1)}}{\sqrt{d}\,F(q)}\frac{F(a)}{F(a-q)}, \qquad 0 \le a < n. \tag{4.2}$$

There is no sample correction: at $a = 0$ the inverse-norm relation gives -1, and at $a = n+q$ the identity $F(n) = 1/\sqrt{d}$ gives -1. All other samples follow directly from the phase formula and the integer sine-shift identities.

For arbitrary integer labels introduce

$$R_p(z) = \frac{F(z)}{F(z-p)}, \qquad K(z) = \frac{\pi e^{-i\omega z}}{\sin \pi z}R_p(z)R_q(z+s), \qquad \omega \equiv \frac{2\pi k}{n} + \pi(p+q-1) \pmod{2\pi},$$

and

$$J_1(z) = \gamma(-z)\gamma(z-p)\gamma(B-z-s)\gamma(A+z+s-q),$$
$$J_2(z) = \gamma(A-z)\gamma(B+z-p)\gamma(-z-s)\gamma(z+s-q).$$

The sine-shift equations give the meromorphic identity

$$K(z+n) - K(z) = -4\pi e^{-i\omega z}\left[(-1)^q \sin\frac{\pi p}{A}J_1(z) + (-1)^s \sin\frac{\pi q}{A}J_2(z)\right]. \tag{4.3}$$

Indeed, put $M_p = R_p(z+n)/R_p(z)$ and $M_q = R_q(z+s+n)/R_q(z+s)$ and split the multiplier into

$$(-1)^{p+q}M_pM_q - 1 = \big((-1)^pM_p - 1\big)(-1)^qM_q + \big((-1)^qM_q - 1\big).$$

Apply the single-ratio sine identity underlying (2.6) to each bracket. The remaining factors are $R_q(z+s+n) = \gamma(B-z-s)\gamma(A+z+s-q)$ and $R_p(z) = \gamma(A-z)\gamma(B+z-p)$; also $\sin\pi(z+s)/\sin\pi z = (-1)^s$.

In the rest of this section assume

$$g, h \geq 1, \qquad p = g + h < n, \qquad 1 \leq l \leq h, \qquad q = -h, \quad s = l - h. \tag{4.4}$$

For any labels define

$$c_0 = \frac{(-1)^{p+q+qs}}{dF(p)F(q)}, \qquad a_0 = (-1)^q \sin\frac{\pi p}{A}, \qquad b_0 = (-1)^s \sin\frac{\pi q}{A}, \qquad I_i = \int_{\mathcal{C}_i} e^{-i\omega z} J_i(z) \frac{dz}{i\sqrt{\Omega}},$$

where $\mathcal{C}_i$ is the meromorphically continued separating contour for its gamma factors.

Lemma 4.1. *For the labels in (4.4), at every discrete frequency,*

$$S_{p,q,s}(k) = -2\sqrt{\Omega}\, c_0(a_0 I_1 + b_0 I_2). \tag{4.5}$$

The same formula holds for

$$(p', q', s') = (-g - l, l, g + h + l)$$

provided its principal frequency satisfies

$$|\omega| < \pi(1 - g/\Omega). \tag{4.6}$$

Proof. For the first assertion let $L = \lfloor p/B \rfloor$ and choose $\sigma = -\varepsilon$ with $0 < \varepsilon < \min(1, (L+1)B - p)$. In $\sigma < \Re z < \sigma + n$ there are no numerator poles of K; $F(z + s - q) = F(z + l)$ has no zero. The noninteger poles are exactly $z_j = p - jB$, $1 \leq j \leq L$. The shifted kernel $K(z + n)$ is regular there. Irrationality of B excludes coincidences with the other divisors.

To turn the vertical integrals into separating integrals one must include the left poles $z_0 = p, z_1, \ldots, z_L$ of J_1 and $z_1, \ldots, z_L$ of J_2. Their other left poles lie to the left of $-l$ and their right poles are nonnegative. The only possibly negative opposite-parameter sums are $-p$ and $B - g - l$. Neither is $-rA - tB$ for nonnegative integers r, t: since $A > n$, such an equality would express a positive integer smaller than n as a positive integer multiple of the irrational number B. Thus the continued contours are nonsingular.

Write $r_{i,j} = \operatorname{Res}_{z=z_j} e^{-i\omega z} J_i(z)$. Equation (4.3) implies, for $j \geq 1$,

$$\operatorname*{Res}_{z=z_j} K(z) = 4\pi(a_0 r_{1,j} + b_0 r_{2,j}).$$

The changes from vertical to separating integrals add respectively

$$\frac{2\pi}{\sqrt{\Omega}} \sum_{j=0}^{L} r_{1,j} \qquad \text{and} \qquad \frac{2\pi}{\sqrt{\Omega}} \sum_{j=1}^{L} r_{2,j}.$$

The residue rectangle consequently gives

$$\sum_{a=0}^{n-1} \operatorname*{Res}_{z=a} K(z) = -2\sqrt{\Omega}(a_0 I_1 + b_0 I_2) + 4\pi a_0 r_{1,0}. \tag{4.7}$$

The horizontal boundaries vanish because $R_p(z)R_q(z + s) = O(e^{-\pi g|\Im z|/\Omega})$ uniformly on the finite strip and $|\omega| \leq \pi$.

The negative column has no patch in this sample interval; the positive column has only its diagonal patch. Hence

$$S_{p,q,s}(k) = c_0 \sum_a \operatorname*{Res}_{z=a} K(z) - B u_q(p + s) e^{-2\pi i k p/n}.$$

On the other hand,

$$r_{1,0} = \frac{\sqrt{\Omega}}{2\pi} e^{-i\omega p}\gamma(-p)R_q(p+s+n), \qquad c_0\, 4\pi a_0 r_{1,0} = Bu_q(p+s)e^{-2\pi ikp/n}.$$

Here one uses $2\sin(\pi p/A)\gamma(-p) = (-1)^p F(p)$ and, at this integer sample, $R_q(p+s+n) = (-1)^q R_q(p+s)$. This proves (4.5).

For the primed labels put $G = g + l$, $H = g + h$, $t = H + l$ and sample the same periodic sum on $a = -t, \dots, n - t - 1$. The raw columns have precisely the patches B at $a = -G$ and $a = -H$, respectively. This follows from the integer sine shifts also when the interval contains negative multiples of n; the numerator arguments avoid positive multiples of n. Choose $\sigma = -t - \varepsilon$ with

$$0 < \varepsilon < \min\{1, (\lfloor h/B \rfloor + 1)B - h, (\lfloor l/B \rfloor + 1)B - l\}.$$

There are no numerator poles. The two noninteger pole families are

$$-G - jB \ (1 \le j \le \lfloor h/B \rfloor), \qquad -H - jB \ (1 \le j \le \lfloor l/B \rfloor).$$

The shifted kernel is regular there. The crossed left poles of J_1 are the first family including $j = 0$; those of J_2 are the first family with $j \ge 1$ and the second family including $j = 0$. Its other left poles are left of σ, and all right poles are right of σ. All noninteger residue contributions cancel by (4.3).

If $h \ne l$, the two surviving integer contributions are

$$Bu_l(h)e^{2\pi ikG/n}, \qquad Bu_{-G}(-H)e^{2\pi ikH/n},$$

and cancel the two patches separately. If $h = l$, some noninteger poles of J_2 and K are double. The same argument uses their full residues, since (4.3) is a meromorphic identity. At the common integer point the first contribution uses the actual second column, with value -1, whereas the second contribution uses the raw first column, with value $1/d$. Their sum is

$$B(1/d - 1)e^{2\pi ikH/n} = (d^{-2} - 1)e^{2\pi ikH/n},$$

exactly the difference between the raw product d^{-2} and the actual product 1. Thus the coincident-diagonal correction is included. Finally $p' + q' = -g$, so the ratio product grows like $O(e^{\pi g|\Im z|/\Omega})$, and the horizontal boundaries vanish under exactly (4.6). □

Write a principal frequency as $\omega = \pi v/n$, with $v \equiv g + 1 \pmod 2$, considered modulo $2n$. Since

$$n(1 - g/\Omega) = n - g + 2g/A, \qquad 0 < 2g/A < 2,$$

condition (4.6) is equivalent to

$$|v| \le n - g. \tag{4.8}$$

It admits $n - g + 1$ frequencies and excludes $g - 1$.

5 The continuous Euler transformation

For separating-contour parameters define

$$I(a, b; c, e; y) = \int e^{2\pi iyz/\Omega}\gamma(a+z)\gamma(b+z)\gamma(c-z)\gamma(e-z)\frac{dz}{i\sqrt{\Omega}}.$$

As usual, after continuation this notation includes all contour residues. Put $\Sigma = a + b + c + e$ and $r = (\Sigma - \Omega)/2$.

Lemma 5.1. *The following meromorphic identity holds:*

$$I(a,b;c,e;y)=\frac{\gamma(a+c)\gamma(b+e)\gamma(a+e)\gamma(b+c)}{\gamma(\Sigma/2+y)\gamma(\Sigma/2-y)}I(a-r,b-r;c-r,e-r;y). \tag{5.1}$$

Proof. Set

$$U=\Omega/4+(r+s)/2, \qquad V=\Omega/4+(r-s)/2,$$
$$K_4(r,s,x,y)=\int e^{2\pi iyz/\Omega}\gamma(U+z)\gamma(U-z)\gamma(V+z+x)\gamma(V-z-x)\frac{dz}{i\sqrt{\Omega}},$$
$$G_r(v)=\gamma(\Omega/2+r+v)\gamma(\Omega/2+r-v).$$

The two-gamma Fourier integral [1, Eq. (2.17)], applied to the two pairs and combined by Fourier convolution, gives

$$K_4(r,s,x,y)=G_r(s)e^{-2\pi ixy/\Omega}K_4(-r,-s,-y,x). \tag{5.2}$$

For precision, the transform of $\gamma(U+z)\gamma(U-z)$ is $\gamma(2U)\gamma(\Omega/2-U+y)\gamma(\Omega/2-U-y)$. Convolution, a translation of its variable by y, and reflection of that variable yield (5.2) with the displayed phase. Interchanging the two positive gamma parameters and translating z by $(s-x)/2$ gives the second identity

$$K_4(r,s,x,y)=e^{\pi iy(s-x)/\Omega}K_4(r,x,s,y). \tag{5.3}$$

Call these operations D and P. The word $DPDPDP$ sends

$$\begin{aligned}(r,s,x,y)&\longmapsto(-r,-s,-y,x)\longmapsto(-r,-y,-s,x)\\&\longmapsto(r,y,-x,-s)\longmapsto(r,-x,y,-s)\\&\longmapsto(-r,x,s,y)\longmapsto(-r,s,x,y).\end{aligned}$$

The six exponential exponents sum to zero. The three Fourier factors multiply to $G_r(s)G_r(x)/G_r(y)$, since $G_{-r}(y)=1/G_r(y)$. Consequently

$$K_4(r,s,x,y)=\frac{G_r(s)G_r(x)}{G_r(y)}K_4(-r,s,x,y). \tag{5.4}$$

These compositions are legitimate before continuation. For example, in the neighborhood $|r|+|s|+|x|+|y|<\Omega/16$, all six four-gamma kernels have positive real parts of their gamma parameters on the imaginary axis and satisfy the strict exponential convergence condition. The convolution is first the ordinary absolutely convergent Fourier convolution for imaginary x,y and nearby real r,s; holomorphy extends it to this neighborhood. All contour translations there cross no pole. The identities then continue meromorphically with their separating contours. Finally centering I by $z\mapsto z+(c-a)/2$ turns (5.4) into (5.1). □

6 A four-term contour identity

Continue to assume (4.4), and put

$$G=g+l, \qquad R=g/2, \qquad X=(n+g)/2+l, \qquad Y=(n+g)/2+h.$$

Lemma 6.1. *With continued separating contours one has*

$$\begin{aligned} 0 = {} & \sin(\pi h/A)K_4(R,X,Y+B,Z) \\ & + (-1)^{G+1}\sin(\pi p/A)e^{-2\pi iZA/\Omega}K_4(R,X,Y-A,Z) \\ & + (-1)^{p}\sin(\pi G/A)e^{-\pi iZA/\Omega}K_4(R,X-A,Y,Z) \\ & - \sin(\pi l/A)e^{-\pi iZB/\Omega}K_4(R,X+B,Y,Z). \end{aligned} \tag{6.1}$$

In the application $|\Re Z| < (\Omega - g)/2$.

Proof. First let $u = \Omega/4 + (R+X)/2$ and $v = \Omega/4 + (R-X)/2$ be general parameters and define

$$\zeta = u - B + z, \quad \eta = u - A - z, \quad \theta = v - B + Y + z, \quad \nu = v - Y - z,$$
$$c = 2u - \Omega, \quad D = v + Y - u, \quad E = c + D, \quad f = u - v - A + Y.$$

Thus $\eta + \zeta = c$, $\eta + \theta = E$, and $\eta - \nu = f$. In this proof abbreviate

$$a(t) = \sin(\pi t/A), \quad b(t) = \sin(\pi t/B), \quad \lambda = e^{2\pi iZB/\Omega},$$

and set

$$\kappa_1 = \frac{a(f-c)}{a(D)}, \quad \kappa_2 = \lambda\frac{b(E)}{b(D)}, \quad \kappa_3 = -\lambda\frac{b(c)}{b(D)}, \quad \kappa_4 = \frac{a(E-f)}{a(D)}.$$

Let $T_i(z)$ be $e^{2\pi iZz/\Omega}$ times the product in row i:

i	gamma product
1	$\gamma(\zeta)\gamma(\eta+\Omega)\gamma(\theta+B)\gamma(\nu)$
2	$\gamma(\zeta+\Omega)\gamma(\eta)\gamma(\theta+B)\gamma(\nu)$
3	$\gamma(\zeta+B)\gamma(\eta)\gamma(\theta+\Omega)\gamma(\nu)$
4	$\gamma(\zeta+B)\gamma(\eta+\Omega)\gamma(\theta)\gamma(\nu)$.

The primitive

$$P(z) = e^{2\pi iZz/\Omega}\gamma(\zeta)\gamma(\theta)\gamma(\eta+\Omega)\gamma(\nu+B)$$

satisfies the exact meromorphic identity

$$\sum_{i=1}^{4}\kappa_i T_i(z) = P(z+B) - P(z). \tag{6.2}$$

To verify it, divide by $-8e^{2\pi iZz/\Omega}\gamma(\zeta)\gamma(\eta)\gamma(\theta)\gamma(\nu)$. The result is the sine identity

$$\begin{aligned} &\kappa_1 a(\eta)b(\eta)a(\theta) + \kappa_2 a(\zeta)b(\zeta)a(\theta) + \kappa_3 a(\zeta)a(\theta)b(\theta) + \kappa_4 a(\zeta)a(\eta)b(\eta) \\ &\qquad = -b(\eta)\{\lambda a(\zeta)a(\theta) + a(\eta)a(\nu)\}. \end{aligned}$$

The coefficient of $\cos(\pi\eta/B)$ vanishes because $\kappa_2 b(c) + \kappa_3 b(E) = 0$. The other coefficient reduces, by sine addition, to $\kappa_1 a(\theta) + \kappa_4 a(\zeta) = -a(\nu)$.

There is a nonempty initial domain where integration of (6.2) has no residue term. Take

$$u = \Omega/2 + \epsilon, \quad v = B/2 + \epsilon, \quad Y = A/2 + \eta_0, \quad \Re z = -n/2, \qquad 0 < \epsilon < A/4, \quad 0 < |\eta_0| < \epsilon.$$

On this line the real parts of ζ, η, θ, ν are $\epsilon, \epsilon, \epsilon + \eta_0, \epsilon - \eta_0$, so every table entry has a separating vertical contour. The first right poles of P have distances $\Omega + \epsilon$ and $B + \epsilon - \eta_0$ from this line, both

greater than B. The shift of its line to the right by B therefore crosses no pole. For $|\Re Z| < (A-4\epsilon)/2$ the horizontal boundaries decay exponentially. Integrating (6.2) gives zero. The strict inequalities persist in an open complex neighborhood, which supplies the domain for meromorphic continuation.

The four table integrals, with measure $dz/(i\sqrt{\Omega})$, equal

$$e^{2\pi iZB/\Omega}K_4(R,X,Y+B,Z), \qquad e^{-2\pi iZA/\Omega}K_4(R,X,Y-A,Z),$$
$$e^{-\pi iZA/\Omega}K_4(R,X-A,Y,Z), \qquad e^{\pi iZB/\Omega}K_4(R,X+B,Y,Z).$$

The respective substitutions in the centered kernels are $z \mapsto z-B, z+A, z+A/2, z-B/2$. Thus the integrated identity continues meromorphically with the specified contours.

At the required parameters,

$$c = G-B, \quad D = h-l, \quad E = p-B, \quad f = p+l-B.$$

If $h > l$, substitute

$$b(p-B) = (-1)^p a(p), \quad b(G-B) = (-1)^G a(G), \quad b(h-l) = (-1)^{h-l+1}a(h-l)$$

and multiply the identity by $e^{-2\pi iZB/\Omega}a(h-l)$. This is (6.1). If $h = l$, then $X = Y$; (5.3) cancels the first and fourth terms and the second and third terms separately, without division by $a(h-l)$.

Finally there is no contour pinch at this specialization. The four lists of opposite-parameter sums are

$$\begin{array}{c|l} 1 & (A+G,\ B-l,\ \Omega+p,\ -h) \\ 2 & (A+G,\ B-l,\ p,\ \Omega-h) \\ 3 & (G,\ \Omega-l,\ A+p,\ B-h) \\ 4 & (\Omega+G,\ -l,\ A+p,\ B-h). \end{array}$$

Their possibly negative entries cannot equal $-rA-tB$ for $r,t \geq 0$, because $A > n$ and B is irrational. These continued values are therefore regular even when poles on the same side merge. □

7 Vanishing of the admitted Fourier coefficients

For the negative column choose the principal $\omega_R \equiv 2\pi j/n + \pi(-g-1) \pmod{2\pi}$. In the range $|\omega_R| < \pi(1-g/\Omega)$ one has

$$\widehat{u}_{-g}(j) = -Be^{ig\omega_R/2}\gamma\left(\frac{\Omega-g}{2}+\frac{\Omega\omega_R}{2\pi}\right)\gamma\left(\frac{\Omega-g}{2}-\frac{\Omega\omega_R}{2\pi}\right). \tag{7.1}$$

Indeed, use the single-column kernel identity with label $-g$ and sample on $0,\dots,n-1$. Formula (4.2) has no patch, the rectangle has no noninteger pole, and $-\epsilon+i\mathbb{R}$, $0<\epsilon<1$, already separates the two gamma pole families. The horizontal boundaries vanish in the stated range. The two-gamma Fourier evaluation and $2\sin(-\pi g/A)\gamma(g) = (-1)^g F(-g)$ yield (7.1).

Proposition 7.1. *Under (4.4), the Fourier transform of $\mathcal{C}_{g,h}(k,l)$ in k vanishes at all $n-g+1$ frequencies in (4.8).*

Proof. Put $\omega_L = -\omega_R$, $Z = \Omega\omega_L/(2\pi)$ and

$$D_0(Z) = \gamma((\Omega-g)/2+Z)\gamma((\Omega-g)/2-Z).$$

The admitted frequencies exclude both endpoint ambiguities, so these choices of principal frequencies agree with the two products in (4.1). For the left product put

$$r = -g/2, \qquad s_1 = -\Omega/2-g/2-h, \qquad s_2 = \Omega/2-g/2-h, \qquad y = -Z.$$

Its two integrals are $e^{\pi ipy/\Omega}K_4(r,s_1,X,y)$ and $e^{\pi i(n+p)y/\Omega}K_4(r,s_2,X,y)$. Apply (5.4), then (5.3), and reflect the integration variable. They become the first and second kernels of (6.1). The two right product integrals are its third and fourth kernels, with centering factors $e^{\pi ip'Z/\Omega}$ and $e^{\pi i(n+p')Z/\Omega}$.

Here are the complete normalizations. After removing the common factor $-2\sqrt{\Omega}$ from the product formulas, the four coefficients in the bracket of (4.1) are

$$M\left(\sin\frac{\pi h}{A},\quad (-1)^{G+1}\sin\frac{\pi p}{A}e^{-2\pi iZA/\Omega},\quad (-1)^{p}\sin\frac{\pi G}{A}e^{-\pi iZA/\Omega},\quad -\sin\frac{\pi l}{A}e^{-\pi iZB/\Omega}\right), \tag{7.2}$$

where

$$M=\frac{B(-1)^{g+h(l+1)}F(G)}{dF(l)}e^{\pi iZ(A+l-g)/\Omega}.$$

To verify the gamma part, the first Euler multiplier is

$$\frac{G_{-g/2}(s_1)G_{-g/2}(X)}{G_{-g/2}(-Z)}=\frac{F(p)F(G)}{F(h)F(l)}\frac{\sin(\pi h/A)}{\sin(\pi p/A)D_0(Z)}.$$

The second multiplier divided by the first is $(-1)^g\sin^2(\pi p/A)/\sin^2(\pi h/A)$. These follow from reflection and

$$\gamma(A+a)=F(-a),\quad \gamma(B-a)=1/F(-a),\quad 2\sin(\pi a/A)\gamma(-a)=(-1)^aF(a).$$

The sample factors are

$$c_L=\frac{(-1)^{g+hl+1}F(h)}{dF(p)},\qquad c_R=\frac{(-1)^{lp+1}F(G)}{dF(l)}.$$

Finally (7.1) reads $\widehat{u}_{-g}(j)=-Be^{-\pi igZ/\Omega}D_0(Z)$, and $\omega_L\equiv -2\pi j/n+\pi(g-1)\pmod{2\pi}$ cancels the remaining integer phases. These substitutions prove (7.2), including all four signs and exponentials. Lemma 6.1 now makes the bracket in (4.1) zero. □

8 From admitted Fourier coefficients to all cubic equations

We record the algebraic normalization needed for the completion argument. Write $\tau=\delta-1$, where $\delta=-d^{-1}$ and $\delta^2=n\delta+1$. The phases constructed above satisfy $W_0=-1$ and $W_aW_{-a}=\delta$ for $a\neq 0$. Their matrix is

$$\mathsf{B}_{a,b}=\delta^{-1}W_aW_{b-a}W_{-b}+\frac{\tau^2}{\delta}\mathbf{1}_{a=b=0},\qquad \mathsf{A}_{a,b}=\frac{\mathsf{B}_{a,b}}{\tau}.$$

Thus $\mathsf{B}_{0,0}=\delta-2$, and every other entry on the two axes or the diagonal is -1. The matrix A has

$$\mathsf{A}_{a,b}=\mathsf{A}_{-b,a-b},\qquad \mathsf{A}_{a,0}=\mathsf{A}_{0,a}=\mathbf{1}_{a=0}-\tau^{-1},\qquad \sum_a\mathsf{A}_{a,0}=-\delta^{-1}. \tag{8.1}$$

All indices in this section are in $\mathbb{Z}/n\mathbb{Z}$ unless integer inequalities are displayed. Define

$$Q(g,h)=\sum_m\mathsf{A}_{m+g,h}\mathsf{A}_{h,m}-\mathbf{1}_{g=0}+\delta^{-1}\mathbf{1}_{h=0}, \tag{8.2}$$

$$C(g,h,k,l)=\sum_m\mathsf{A}_{m,g+h}\mathsf{A}_{g,m+k}\mathsf{A}_{h,m+l}-\mathsf{A}_{g+l,k}\mathsf{A}_{h+k,l}+\delta^{-1}\mathbf{1}_{g=h=0}, \tag{8.3}$$

$$\mathcal{C}_{g,h}(k,l)=\tau^3C(g,h,k,l). \tag{8.4}$$

The previously established quadratic identities say $Q(g,h)=0$ for every g,h. The admitted Fourier result in Proposition 7.1 concerns the last normalization.

8.1 Balanced triples and collisions

For a cubic index quadruple put

$$\boldsymbol{a} = (0, k-g, l-h), \qquad \boldsymbol{c} = (g+h, -k, -l). \tag{8.5}$$

Their six entries sum to zero. The cross sums are

$$(a_i + c_j)_{i,j=0}^{2} = \begin{pmatrix} g+h & -k & -l \\ h+k & -g & k-g-l \\ g+l & l-h-k & -h \end{pmatrix}. \tag{8.6}$$

Conversely, every balanced pair of triples has this form after opposite translation: replace a_i by $a_i - a_0$ and c_j by $c_j + a_0$, then take $k = -c_1$, $l = -c_2$, $g = -c_1 - a_1$, and $h = -c_2 - a_2$. Call a pair *cross-free* if none of its nine cross sums is zero.

Lemma 8.1. *Every cubic with a zero in (8.6) follows from the quadratic identities and the phase inverse norms. For a cross-free pair define*

$$S_W(\boldsymbol{a}, \boldsymbol{c}) = \sum_m \prod_{i=0}^{2} W_{m+a_i} \prod_{j=0}^{2} W_{c_j - m}, \qquad P_W(\boldsymbol{a}, \boldsymbol{c}) = \prod_{i,j=0}^{2} W_{a_i + c_j}.$$

Then

$$\mathcal{C}_{g,h}(k,l) = \frac{W_{-g-h} W_g W_h}{\delta^5} \left\{ \delta^2 S_W(\boldsymbol{a}, \boldsymbol{c}) - \tau P_W(\boldsymbol{a}, \boldsymbol{c}) \right\}. \tag{8.7}$$

Consequently, vanishing of a cross-free cubic is preserved by independent permutations and opposite translation of its triples.

Proof. Choose either square root s of δ and put $b_a = W_a/s$, $\lambda = s/\tau$, and $\mu = \tau/\delta$. These are auxiliary algebraic variables, with no complex conjugation in their definition. They satisfy

$$b_0 = -s^{-1}, \qquad b_a b_{-a} = 1 \ (a \neq 0), \qquad \mathsf{A}_{u,v} = \lambda b_u b_{v-u} b_{-v} + \mu \mathbf{1}_{u=v=0}.$$

Write S_b and P_b for the corresponding six-factor sum and nine-factor product. If $g, h, g+h \neq 0$, direct substitution in (8.3) gives

$$C(g,h,k,l) = \lambda^3 b_{-g-h} b_g b_h \left(S_b - \lambda^{-1} P_b - \frac{\tau^3}{\delta^2} \mathbf{1}_{\{c_0,c_1,c_2\} = \{-a_0,-a_1,-a_2\}} \right). \tag{8.8}$$

Here equality inside the indicator means equality of multisets. Indeed, an origin correction on the right side of the cubic can occur only at $(k,l) = (0,-g)$ or $(-h,0)$, and its normalized coefficient is $\mu/\lambda^2 = \tau^3/\delta^2$. These are exactly the two possible multiset matches. Under the three nonzero conditions on $g, h, g+h$, every such match has three distinct row entries: a matching permutation must avoid the three nonzero diagonal cross sums, leaving just the two derangements, and a repeated row would also permit a forbidden match.

For a full match with distinct rows, inverse norms alone give

$$S_b = n - 3 + 3\delta^{-1}, \qquad \lambda^{-1} P_b = -\tau \delta^{-2}.$$

Their difference is τ^3/δ^2, using $n = \delta - \delta^{-1}$. For a collision without a full match, permute so that $a_0 + c_0 = 0$. Then

$$b_{m+a_0} b_{-m-a_0} = 1 + (\delta^{-1} - 1) \mathbf{1}_{m=-a_0}.$$

The unrestricted sum of the other four factors is, after reindexing,

$$\sum_t b_t b_{H-t} b_{-t-V} b_{t+V-H}, \qquad H = a_1 + c_1, \quad V = a_2 + c_1.$$

Both H and V are nonzero, since otherwise the remaining two pairs also match. This sum is $\lambda^{-2} Q(-V, H)$ and hence vanishes. The surviving contribution to S_b is

$$(\delta^{-1} - 1) \prod_{i=1}^{2} b_{a_i - a_0} \prod_{j=1}^{2} b_{c_j + a_0}.$$

The four interior factors of P_b pair to one; thus P_b is b_0 times the displayed boundary product. Since $\lambda^{-1} b_0 = \delta^{-1} - 1$, this proves the collision case of (8.8).

For completeness, the boundary cases $g = 0$, $h = 0$, or $g + h = 0$ can be checked without dividing by the preceding prefactor. Set $c = \tau^{-1}$. For $g = 0$, $h \neq 0$, the cubic sum reduces by $Q = 0$ to

$$\mathsf{A}_{-k,h} \mathsf{A}_{h,l-k} - c \mathbf{1}_{l=0}.$$

For $l \neq 0$ the phase formula identifies this with $\mathsf{A}_{l,k} \mathsf{A}_{h+k,l}$. For $l = 0$ use

$$\mathsf{A}_{u,h} \mathsf{A}_{h,u} = \begin{cases} c^2, & u = 0 \text{ or } u = h, \\ \lambda^2, & u \neq 0, h, \end{cases} \qquad (h \neq 0), \qquad \lambda^2 - c = c^2.$$

The case $h = 0$, $g \neq 0$ follows by interchanging (g, k) and (h, l). For $h = -g \neq 0$ the cubic sum reduces to

$$\mathsf{A}_{g,k} \mathsf{A}_{-l,g} - c \mathbf{1}_{k=g+l}.$$

Again the phase formula handles $k \neq g + l$, and the two product values above handle $k = g + l$. Finally, for $g = h = 0$ the cubic sum is

$$\mathbf{1}_{k=l=0} - c(\mathbf{1}_{k=0} + \mathbf{1}_{l=0} + \mathbf{1}_{k=l}) + 3c^2 - nc^3.$$

It equals $\mathsf{A}_{l,k} \mathsf{A}_{k,l} - \delta^{-1}$: the respective product values are $(1-c)^2$, c^2, and λ^2, according as $k = l = 0$, exactly one of $k = 0, l = 0, k = l$ holds, or none holds. The dimension equation verifies all three equalities. This proves all boundary cubics.

In the cross-free case there is no origin correction in any factor of the cubic. Substitution of the phase formula and $W_a W_{-a} = \delta$ gives (8.7); equivalently it follows from (8.8) using $S_W = s^6 S_b$ and $P_W = s^9 P_b$. Its prefactor is nonzero. Both S_W and P_W are invariant under independent permutations, and opposite translation is absorbed by reindexing m in S_W. □

8.2 An induction on the first index

Proposition 8.2. *At every odd $n \geq 3$, all ordered cubics vanish:*

$$\mathcal{C}_{g,h}(k, l) = 0 \quad (g, h \geq 1,\ g + h < n,\ 1 \leq l \leq h,\ k \in \mathbb{Z}/n\mathbb{Z}).$$

Proof. Fix n and induct on the integer g. By Proposition 7.1, for each ordered g, h, l, precisely $n - g + 1$ of the n Fourier coefficients of the sequence $k \mapsto \mathcal{C}_{g,h}(k, l)$ are known to vanish. The omitted frequencies form a consecutive cyclic block of length $g - 1$: the principal integers $v = n\omega_R/\pi$, of parity $g + 1$, advance by two as the Fourier index advances by one, and the admitted condition is

$|v| \le n-g$. For $g=1$ there are no omitted frequencies, proving the base case. For $g>1$, Fourier inversion therefore gives, with $\zeta = \exp(2\pi i/n)$,

$$\mathcal{C}_{g,h}(k,l) = \zeta^{ak} P(\zeta^k), \qquad \deg P \le g-2 \tag{8.9}$$

for an integer a and a complex polynomial P.

We show that $k = 1, \dots, g-1$ are zeros. A collision is already covered by Lemma 8.1. Otherwise perform the following changes of cubic labels:

$$(g,h,k,l) \longmapsto \begin{cases} (k,h,g,l+g-k), & l+g-k \le h, \\ (k,n-h-k,k+h-l,n-g-l), & l+g-k > h. \end{cases} \tag{8.10}$$

These changes arise from permutations and opposite translation of (8.5). With positions numbered $0,1,2$, the first uses rows $(1,0,2)$ and unchanged columns; the second uses rows $(2,0,1)$ and columns $(2,1,0)$. In both cases translate the first row entry to zero. Substitution gives exactly (8.10), with all entries interpreted modulo n where necessary.

The new first index is $k < g$. Both new quadruples are ordered. This is immediate in the first case; in the second,

$$k + (n-h-k) = n-h < n, \qquad 1 \le n-g-l \le n-h-k,$$

where the last inequality follows from $l+g-k > h$. The induction hypothesis and the nonzero factor in (8.7) therefore give each required zero of the original cubic. The $g-1$ points $\zeta, \dots, \zeta^{g-1}$ are distinct, including for composite n, so (8.9) forces $P = 0$. □

8.3 A signed normal form

Lemma 8.3. *Every cross-free balanced pair of triples can be put into the ordered range of Proposition 8.2 by independent permutations and opposite translation, possibly after negating all six entries simultaneously.*

Proof. Let $X_{ij} \in \{1, \dots, n-1\}$ be the integer representative of $a_i + c_j$. Balance implies $X_{00} + X_{11} + X_{22} \in \{n, 2n\}$. Negation replaces each X_{ij} by $n - X_{ij}$, so we may arrange that the diagonal sum is $2n$.

There exist $i \ne j$ with $X_{ij} \ge X_{jj}$. Otherwise each diagonal entry would be a strict maximum in its column. The three row labels would then be distinct. Let $d_j > 0$ be the forward cyclic gap from a_j to the next row label. Since the other entry in column j is smaller and no cross sum is zero, $X_{jj} + d_j > n$. The gaps sum to n, so summing would give $\sum_j X_{jj} > 2n$, a contradiction.

Choose such i, j, let r be the remaining position, and use row and column order (i, r, j). Normalize the first row to zero. The resulting cubic indices have

$$g = n - X_{rr}, \qquad h = n - X_{jj}, \qquad l = n - X_{ij}.$$

Thus $g, h \ge 1$, $1 \le l \le h$, and $g + h = 2n - X_{rr} - X_{jj} = X_{ii} < n$, as required. □

9 Reconstruction and infinitude

Suppose $2^m \equiv -1 \pmod{n}$ for some $m \ge 1$. The characteristic-zero Frobenius automorphism established in Proposition 3.1 fixes δ and sends W_a to W_{2a}. Consequently its mth power sends

$$W_a \longmapsto W_{-a}, \qquad \mathsf{B}_{a,b} \longmapsto \mathsf{B}_{-a,-b}.$$

Applying it to a cubic and replacing the summation index by its negative shows that simultaneous negation of all four cubic indices preserves vanishing. Lemma 8.3, Proposition 8.2, and Lemma 8.1 now prove $C(g,h,k,l)=0$ for every quadruple.

We explicitly check that these equations imply all hypotheses of Evans–Gannon reconstruction. In addition to (8.1) and $Q=0$, their equations require the quartic residual

$$K(g,h,i,k)=\sum_{l,m}\mathsf{A}_{l,m}\mathsf{A}_{l+g,h}\mathsf{A}_{h+m,l+i}\mathsf{A}_{i,k+m}-\mathsf{A}_{h-g,i-g}\mathbf{1}_{k=g}\\ +\delta^{-1}\mathbf{1}_{h=0}\mathsf{A}_{i,k}+\delta^{-1}\mathsf{A}_{g,h}\mathbf{1}_{i=0}$$

to vanish. Order-three symmetry alone gives the polynomial identity

$$K(g,h,i,k)=\sum_{m}\mathsf{A}_{i,k+m}C(-h,h+m,g-h,i)+\mathsf{A}_{i-h,g-h}Q(g-k,i). \tag{9.1}$$

Indeed, replace $\mathsf{A}_{l+g,h}$ by $\mathsf{A}_{-h,l+g-h}$ in the double sum. The inner sum is

$$C(-h,h+m,g-h,i)+\mathsf{A}_{i-h,g-h}\mathsf{A}_{m+g,i}-\delta^{-1}\mathbf{1}_{h=0}\mathbf{1}_{m=0}.$$

Multiplication by $\mathsf{A}_{i,k+m}$ and summation over m leave the quadratic sum $Q(g-k,i)+\mathbf{1}_{g=k}-\delta^{-1}\mathbf{1}_{i=0}$. The remaining terms cancel using $\mathsf{A}_{i-h,g-h}=\mathsf{A}_{h-g,i-g}$ and, when $i=0$, $\mathsf{A}_{-h,g-h}=\mathsf{A}_{g,h}$. This proves (9.1), hence $K=0$.

Equations (8.1), (8.2), and (9.1) are exactly equations (4.7)–(4.10) of [2] with $\omega=1$ and the negative choice $\delta=(n-\sqrt{n^2+4})/2$. Theorem 2(a) of [2] allows this choice. It constructs a spherical fusion category with simple objects α_a and $\alpha_a\rho$, $a\in\mathbb{Z}_n$, and fusion rules

$$\alpha_a\alpha_b=\alpha_{a+b},\qquad \alpha_a(\alpha_b\rho)=\alpha_{a+b}\rho,\qquad (\alpha_b\rho)\alpha_a=\alpha_{b-a}\rho,\\ (\alpha_a\rho)(\alpha_b\rho)=\alpha_{a-b}+\sum_{c\in\mathbb{Z}_n}\alpha_c\rho.$$

The categorical dimensions are 1 and δ.

The entries are algebraic by Lemma 2.2. An embedding of $\mathbb{Q}(\delta)$ taking δ to $d=(n+\sqrt{n^2+4})/2$ extends to the number field of matrix coefficients. Apply this embedding to the reconstruction equations and reconstruct again. The resulting spherical category has categorical dimensions 1 and d, which are the Frobenius–Perron dimensions of the displayed fusion rules. Its global dimension is therefore $n(1+d^2)$, equal to its Frobenius–Perron dimension, so it is pseudo-unitary. This conclusion does not assert unitarity: Theorem 2(c) of [2] additionally requires the positively conjugated matrix to be Hermitian, a property not established here.

Finally, $n=2^m+1$ is odd and $2^m\equiv -1 \pmod{n}$ for every $m\geq 1$. These orders are unbounded and the resulting categories have distinct ranks $2n$. Hence there are infinitely many complex spherical cyclic Haagerup–Izumi fusion categories, and infinitely many pseudo-unitary such categorifications.

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